

CAVYN

Autonomous Multi-Agent Voice Boardroom

Next-generation conversational arbitration engine evaluating venture pitches across Strategic Scale (CEO), Technical Architecture (CTO), and Capital Viability (CFO) with sub-15ms acoustic interruption and monotonic epoch state validation.

Zero-Latency Barge-In <15ms local silence with zero audio leaks	3 Autonomous Brains Elena (CEO), Marcus (CTO), Vikram (CFO)	Deterministic Arbiter Monotonic epoch validation and priority matrix	Zero-Key Deployment Runs instantly on Vercel with optional BYOK
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Bridging Founders and Institutional Advisory

THE FOUNDER DILEMMA

- **Intimidating Pitch Dynamics**

First-time builders face stressful, adversarial viva-style question rounds without empathetic feedback.

- **Technical & Financial Jargon**

Founders stumble on wall-street acronyms (CAC payback, LTV, P99 SLA) rather than exploring true customer value.

- **Rigid AI Limitations**

Traditional chatbot simulations sound mechanical, talk down to users, and cannot handle live acoustic interruption.

THE CAVYN SOLUTION

- **Collaborative 3-Seat Mentorship**

Elena (CEO), Marcus (CTO), and Vikram (CFO) act as empathetic peers advising you casually over coffee.

- **Sub-15ms Acoustic Barge-In**

Speak at any moment—active mentor audio silences locally in <15ms with guaranteed zero stale audio leaks.

- **Real-Time Discrepancy Ledger**

Automatically logs claims into Market, Tech, and Financial buckets, detecting and flagging metric contradictions live.

Three Specialized Brains in Continuous Dialogue

SEAT 01 // ELENA VANCE

Lead Partner & CEO

Focus Domain:

Market Scale, Moat & Narrative

Voice & Mentoring Style:

Warm, visionary, supportive. Asks about early users, customer pain points, and distribution velocity.

Example Inquiries:

- "Who was the first person to try your app?"
- "What is the biggest pain you solve for them?"

SEAT 02 // MARCUS STONE

Technical Advisor & CTO

Focus Domain:

Architecture, Moat & Scalability

Voice & Mentoring Style:

Curious developer teammate on Discord. Avoids alienating jargon; focuses on snappy user experience.

Example Inquiries:

- "What frameworks are you building with?"
- "What was your hardest technical hurdle?"

SEAT 03 // VIKRAM RAO

Financial Mentor & CFO

Focus Domain:

Unit Economics & Cash Health

Voice & Mentoring Style:

Calm, encouraging financial advisor. Breaks down pricing models and runway into simple concepts.

Example Inquiries:

- "How will you charge paying customers?"
- "Where will your initial capital be deployed?"

Sub-15ms Acoustic Interruption & Turn Arbitration

BARGE-IN LATENCY

< 15ms

Instant local silence

STALE AUDIO LEAKS

0 Frames

Strict epoch invalidation

MONOTONIC EPOCHS

100% Sync

No dropped turns

SUBTITLE STREAM

Dual-Party

Persistent transcript tray

FOUR-STAGE CONVERSATIONAL TURN LIFECYCLE

01. Acoustic Detection

Web Speech captures speech interim frames. Playing agent audio is muted locally in <15ms without server network roundtrip.

02. Parallel Bid Evaluation

Final utterance streams to Google Gemini 2.5 Flash. All three committee seats evaluate turn bids with urgency scores (1-100).

03. Deterministic Arbiter

Cavyn arbiter evaluates addressed roles, contradiction interrupts, and conversational fairness to elect the single speaker.

04. Synthesis & Broadcast

Speech is synthesized via natural human voices with monotonic epoch tagging, ensuring zero playback if interrupted.

Cyber-Institutional HUD & Radial Sector Navigation

GTA 5 Radial Hub (Press Q)

Interactive 360° circular pie-sector wheel allowing instant switching between Chamber, Claims Dossier, Founder Profile, Telemetry, and Voting Matrix with authentic reticle HUD graphics.

Live Discrepancy & Claims Ledger

Categorizes spoken assertions into Financial, Technical, and Market buckets. Flags conflicting numbers in real-time with visual conflict indicators and audit timestamps.

Interactive SAFE Dilution Calculator

Dynamic slider modeling for investment tranches (\$100k-\$2M) against post-money valuation caps (\$2M-\$25M) with live pro-forma ownership split bars.

Persistent Subtitle & Audio Stream

Live high-frequency 12-bar responsive audio spectrum displaying real-time founder utterances and persistent mentor replies that never vanish abruptly.

Enterprise Architecture with Zero-Key Deployment

FRONTEND ENGINE

Next.js 16 (App Router)

Turbopack, React 19, Tailwind CSS v4,
Lucide Icons

MULTI-AGENT BRAIN

Google Gemini 2.5 Flash

Structured JSON responses, Search
grounding & role prompts

VOICE SYNTHESIS

Dual Audio Pipeline

Natural browser human voices (Jenny,
Guy, Ryan) + Rime VOX

STATE & RUNTIME

Thread-Safe Orchestrator

In-memory state machine with
monotonic epoch tracking

SEAMLESS VERCEL CLOUD DEPLOYMENT (ZERO KEYS REQUIRED)

Cavyn requires ZERO mandatory environment variables to build and deploy. Anyone can deploy to Vercel with 1 click. Visitors can test immediately using the built-in offline simulation model, or plug in their personal Google Gemini key directly in the web UI for real-time live LLM deliberation with in-memory encryption.

- Live URL: <https://cavyn-boardroom.vercel.app>
- GitHub: <https://github.com/Nitanshu715/Cavyn>
- Full offline fallback for air-gapped or keyless execution

Cavyn: The Future of Autonomous Founder Mentorship

Key Achievements & Capabilities:

- ' Real-Time Multi-Agent Debate: Elena, Marcus, and Vikram converse and cross-examine like an authentic 4-person board.
- ' Ultra-Fast Acoustic Interruption: Sub-15ms local muting allows founders to speak naturally without waiting for awkward turn timeouts.
- ' Relaxed Human Dialogue: Supportive peer coaching completely eliminates stressful viva/interrogation vibes.
- ' Institutional Due Diligence: Automated claim extraction, contradiction detection, and interactive SAFE dilution modeling.
- ' Zero-Key Cloud Deployment: Runs smoothly on Vercel out of the box with zero mandatory secrets and optional runtime BYOK.

Production Deployment: <https://cavyn-boardroom.vercel.app>

GitHub Repository: <https://github.com/Nitanshu715/Cavyn>